

# KRITIK SOMAN

Seeking opportunities in Applied AI/ML Research and Engineering | 5+ yrs. experience

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## PROFESSIONAL EXPERIENCE

### Accenture

Bengaluru, India

AI/ML Computational Science Associate Manager (Sept. 2025 to Present)

AI/ML Computational Science Specialist (April 2022 to Aug. 2025)

#### ApprovalX: Intelligent Compliance Automation Platform

- Developed a **multi-lingual agentic AI** system in **python** using **LangGraph** and **GPT-4o** to automate review workflows for marketing assets like emails and banners. Built an **orchestration agent** to route inputs to relevant detectors based on textual input from user using LLM. Implemented a **branding agent** with LLM tools that validates content against database-driven brand guidelines. Designed a **functional agent** to compare uploaded assets with user-provided reference files. Created a **compliance agent** for automated medical-legal document review. Engineered a **reporting agent** that consolidates errors and generates targeted remediation suggestions with LLM. Deployed with **Ray serve**, **FastAPI**, **Kafka**, and **Docker**.
- Engineered a multi-lingual font verification pipeline for e-commerce banners using **HiSAM**, **Google OCR**, and **glyph matching**—accurate down to 10px font size and low-contrast text.
- Performed **SFT**, **PEFT** on **Qwen 2.5-VL** for performing various compliance checks on documents.
- Delivered a **30% reduction in Average Handling Time (AHT)** for manual QA through automation and intelligent document parsing.

#### InstructGen: GenAI-Powered Marketing Automation Suite

- Built a **Retrieval-Augmented Generation (RAG)** pipeline in python for brand specific email **HTML generation** with custom CSS.
- Developed a **RAG-based SQL generation** engine in python for customer targeting by merging sub-queries indexed in a vector database.
- Achieved a **50% reduction in AHT** for marketing email creation and campaign targeting through end-to-end automation.

#### Xtracto: Cross-Format Document Parsing Engine

- Architected a standardized parser in python for diverse digital formats (PPTX, PSD, PDF, HTML, raster images), integrating tools like pdfminer, psd-tools, Aspose API, Playwright, pdf2image, Google OCR, Microsoft Florence VLM, and HiSAM for LLM-based automation.

#### Leadership & Mentorship:

- Leading a team of 4 AI Analysts through full-cycle development, driving iterative improvements and production readiness.
- Conducted hiring and technical interviews for entry-level AI Analyst roles, contributing to team growth and talent acquisition.

### App Orchid

Hyderabad, India

Junior Data Scientist (August 2020 to April 2022)

- NLP**: Created methods for extraction of custom entities in contracts based on Question Answering (QA), RegEx, Shallow Syntax Parsing, Token REGEX, Relative Time Expression, Language Identification, Fine tuning QA, Sentence similarity models, Classification of Document, Clauses.
- CV**: Row and column detection in table images using cascade RCNN. Heading, text, table detection in contract images using YOLO v3+.

### STMicroelectronics

Greater Noida, India

Intern (February 2016 – June 2016)

Undergraduate thesis on **Motion Algorithms using MEMS Sensors**.

## PATENTS / PUBLICATIONS

- Method and System for Modifying Images (US Patent Filed)
- Artificial Intelligence based System and Method for Document Analysis (US Patent Filed)
- Soman, Kritik. "Rooftop Detection using Aerial Drone Imagery." In Proceedings of the ACM India Joint International Conference on Data Science and Management of Data, pp. 281-284. 2019.

## EDUCATION

M.Tech. in *Machine Learning*

Indian Institute of Technology, Kanpur

August 2018 – June 2020 | GPA: 9.3/10

- Thesis on **Semantic Mobile Base Station**

**Placement:** Developed a method using Genetic Algorithm and Semantic Segmentation for placement of mobile base stations at optimal location given RGB and depth aerial images. (Supervisor: Prof. K.S. Venkatesh)

B.Tech. in *ECE* – IIIT Noida

August 2012 – June 2016 | GPA: 9/10

## AWARDS

Accenture Technology Wizard Award 2025  
| Excellent Mentor 2023 | Client Value Creation Award 2023 | IIT Kanpur Academic Excellence Award for 2018-19 |  
ACM CoDS-COMAD Travel Grant 2019, 2020

## OPEN SOURCE

- GIMP-ML** - Deep Learning Models for GNU Image Manipulation Program (1.5k+ ★ on GitHub), News: Visual Informatics Group @ UT Austin, Papers With Code (Top Trending), Presented at PyCon India 2024 @Google

- VoiceFlow** : A voice powered GUI automation tool using agents (**part of Berkeley Advanced LLM MOOC – Ninja Tier**)

## FRAMEWORKS

Python | **AI Agents** | **LangGraph** | **RAG** | **LLMs** | Google Agent Development Kit | **PyTorch** | **Transformers** | Spacy | OpenCV | skimage | Pandas | playwright | psd tools | Postgres | FastAPI | pdfminer | Git